

Sampling-Boundary Distortion in Sharpe-Ratio Evidence from Financial Panels

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Abstract

Entity-time financial panels can be mis-evaluated when selected rows are treated as independent observations, even when the economically meaningful object is a date-level portfolio return. This paper studies the resulting sampling-boundary distortion for Sharpe-ratio evidence. The central diagnostic compares the long-run date-level delta-method variance of the Sharpe estimator with the row-pooled IID variance that would be reported under independent selected rows, termed here the Sharpe Unified Variance Inflation Factor. The diagnostic is paired with date aggregation and date-level HAC/bootstrap inference. Researcher-menu correction, placebo tests, factor-alpha checks, and turnover-cost sensitivity are used as claim-boundary diagnostics. Public Kenneth French panel benchmarks and AQR single-series factor benchmarks illustrate composite panel-audit and time-series-only cases. In null designs with cross-sectional and serial dependence, row-naïve rejection reaches about 35 percent at a nominal 5 percent level, while date-level HAC and bootstrap procedures remain closer to nominal size. The contribution is a target-aware distortion record for the row-to-date target change that standard clustering, Sharpe time-series inference, and menu correction address only separately.

Keywords: Sharpe ratio; sampling-boundary distortion; financial econometrics; panel data; cross-sectional dependence; serial dependence; HAC inference; dependent bootstrap; multiple testing; portfolio signal evaluation

JEL Codes: C12; C15; G11; G12; G17

1 Introduction

1.1 Problem: The wrong sampling boundary

Empirical finance evaluates signals in entity-time panels: factor screens, anomaly sorts, automated portfolio-selection pipelines, model-selection grids, and trading-strategy back-tests. A typical panel can be written as

$$\mathcal{P} = \{(t, i, s_{t,i}, r_{t+1,i})\},$$

where t indexes dates, i indexes assets or portfolios, $s_{t,i}$ is a signal, and $r_{t+1,i}$ is the next-period return. One evaluation failure mode is to evaluate a positive edge using all selected symbol-date rows as though they were independent. This turns a panel with T dates and n_t selected rows per date into a nominal sample of $\sum_t n_t$, even when same-date rows share common shocks. This row-level calculation is used here as a baseline reference for bias quantification and as a concrete failure mode for automated signal-evaluation pipelines. The failure is not merely a standard-error choice; it changes the sampling unit from dates to rows. This baseline is not presented as best-practice empirical asset pricing; it is the default failure mode in automated or non-clustered signal screens.

The issue extends beyond a single trading-strategy application. Factor screens, anomaly panels, model-selection procedures, and automated portfolio-signal evaluations can all multiply observations across assets on the same date while the economic payoff is realized through a date-level portfolio return. The paper uses public trading-panel and factor-screen examples because they make the sampling boundary visible, but the statistical problem is a general financial-econometrics question: what uncertainty should be attached to a Sharpe estimate after cross-sectional row multiplication and serial dependence in the date-return series are recognized. The paper is therefore a framework paper illustrated with public benchmark panels, not a comprehensive survey of all finance-panel evidence.

The problem is not only a matter of notation. A row-pooled t -test implicitly treats the cross-section on a date as many independent draws from the same experiment. A portfolio signal or trading strategy, however, is marked to market by date. If an evaluation selects several assets or portfolios on the same date, the economically relevant object is the date-level portfolio return after aggregation. Cross-sectional dependence determines how much extra information the additional selected rows actually contain, and serial dependence determines the long-run variance of the date series. Standard clustering techniques, including the finance-panel guidance of Petersen (2009), address standard errors for linear regression coefficients or means. They do not by themselves report the distortion between a row-pooled Sharpe claim and the long-run variance of the economically relevant date-level Sharpe estimator.

Automated signal-search pipelines can evaluate large numbers of entity-date rows and many menu variants in a single run. The computational ease of row-level evaluation makes the sampling-boundary error easier to automate than to detect. The threat model is therefore concrete: the problem is largest when multiple same-date entities are selected, returns share common shocks, the performance statistic is nonlinear, the researcher searches over menus, date returns have serial dependence, and trading or implementation costs are nontrivial when the result is interpreted as tradable performance.

The object that records this boundary change is the Sharpe Unified Variance Inflation Factor, or Sharpe UVIF:

$$\text{UVIF}_{\text{SR}} = \frac{\text{Var}_{LR}(\widehat{\text{SR}}_{\text{date}})}{\text{Var}_{\text{IID}, \text{row}}(\widehat{\text{SR}}_{\text{row}})}.$$

It is best interpreted as a sampling-boundary distortion ratio for reported Sharpe uncertainty. The numerator is the uncertainty of the economically relevant date-level Sharpe claim. The denominator is the uncertainty that would be reported by a row-pooled IID analysis of the selected entity-date returns. When the row-pooled and date-level estimators target the same Sharpe functional under exchangeability and equal-weight aggregation, Sharpe UVIF reduces to a design-effect variance inflation factor for the nonlinear Sharpe ratio. More generally, it is a target-aware diagnostic bridge: date-level HAC inference supplies the confirmatory date-level uncertainty estimate, while Sharpe UVIF shows how much of the original row-pooled precision came from the wrong sampling boundary.

1.2 Solution: Dependence-aware inference

This paper makes three contributions. First, it formalizes sampling-boundary distortion for Sharpe-ratio evidence by comparing row-pooled reported uncertainty with date-level long-run uncertainty; the resulting diagnostic is Sharpe UVIF. Unlike mean-only design effects, Sharpe UVIF applies the delta method to the nonlinear Sharpe functional and therefore carries the long-run covariance of both returns and squared returns into the uncertainty calculation. Second, it embeds this variance accounting in a pre-specified evaluation design that separates cross-sectional row multiplication from serial dependence in date-level portfolio returns. Third, it uses public benchmark panels to show the interpretive consequences of the boundary change and Monte Carlo null designs to document the size distortion created by row-naïve inference.

The empirical section validates the pipeline on canonical French WML, evaluates French panel stress cases, and reports AQR factor benchmarks as time-series-only evidence. The public Kenneth French and AQR data are rendered into the manuscript from production CSV artifacts. The conservative empirical question is whether apparently significant portfolio-signal claims are artifacts of row-level sampling and disappear after

date aggregation, dependence adjustment, and researcher-menu correction. The same code also runs a Monte Carlo study comparing row-naive testing with date-level bootstrap, HAC-delta, block bootstrap, stationary bootstrap, and joint resampling decisions.

2 Related work

This paper sits at the intersection of literatures that are often discussed separately. Survey and grouped-error design effects explain why row counts overstate information (Kish, 1965; Moulton, 1986), and finance-panel standard error work explains why cross-sectional and time clustering matter in empirical finance (Petersen, 2009). Those results are central background, but they mainly address means or regression coefficients rather than nonlinear portfolio-performance evidence after entity-date rows have been aggregated into a date-level return. They do not define a target-aware distortion ratio for nonlinear portfolio performance statistics.

Sharpe-ratio inference follows a separate time-series literature. Lo (2002) studies Sharpe-ratio sampling under return dynamics, and robust Sharpe testing builds on HAC and bootstrap methods (Ledoit and Wolf, 2008; Opdyke, 2007; Andrews, 1991; Kiefer and Vogelsang, 2005; Kunsch, 1989; Politis and Romano, 1994; Lahiri, 2003). These tools define the date-level sampling law once the return series is specified. They do not quantify how much apparent Sharpe precision was created before the evidence was moved from selected entity-date rows to the economically relevant date-return boundary. The boundary problem occurs before Sharpe inference: which return series is the object?

Researcher-menu and factor-zoo work addresses a third source of false precision. White Reality Check and Romano-Wolf procedures control data-snooping and familywise error in menus (White, 2000; Romano and Wolf, 2005), while the empirical asset-pricing literature documents false discoveries, factor-alpha fragility, anomaly decay, and trading costs (Harvey et al., 2016; Harvey and Liu, 2020a,b; Bailey and Lopez de Prado, 2014; Holm, 1979; Fama and French, 1993; Carhart, 1997; Fama and French, 2010; McLean and Pontiff, 2016; Almgren and Chriss, 2001; Hasbrouck, 2009; Frazzini et al., 2015). This paper is adjacent to that literature but distinct: its object is the sampling unit and return boundary before a row-level Sharpe claim is interpreted as portfolio-performance evidence. It separates three failure channels that are often bundled together in practice: cross-sectional row multiplication, serial dependence in the date-return series, and researcher-menu search.

Recent replication work in empirical asset pricing asks whether published predictors survive stricter implementation, data, and multiple-testing standards. The present paper asks an upstream question: before a predictor is interpreted as portfolio-performance evidence, has the sampling unit been moved from selected entity-date rows to the date-level return that can actually be allocated to?

3 Sampling-boundary framework

The framework is a dependence-aware inference design rather than a single test. Its first channel is cross-sectional: all evidence is translated from selected symbol-date rows into one economically interpretable date-level portfolio return, so same-date clusters cannot create sample size by row multiplication. Its second channel is serial: all Sharpe decisions are made on the date-return series using HAC-delta and dependent-bootstrap procedures. For panel candidates with row-level signal-return pairs, a claim must survive both channels before the paper asks the separate researcher-menu, placebo, factor-alpha, and cost questions.

Each panel candidate is reported at three levels: the row-naive baseline, the date-level confirmatory estimate, and the UVIF/ERIR diagnostic gap between them.

3.1 Date aggregation as the target boundary

Let S_t denote the selected entities on date t , let $Y_{t+1,i}$ be the signed one-period row return, and let $w_{t,i}$ be the economic weight assigned to selected row i . The portfolio return used for Sharpe inference is the date-level object

$$R_{t+1} = \sum_{i \in S_t} w_{t,i} Y_{t+1,i}, \quad \sum_{i \in S_t} |w_{t,i}| < \infty. \quad (1)$$

This is the target boundary. A row-pooled analysis estimates uncertainty as though the selected $Y_{t+1,i}$ were independent observations, while the confirmatory analysis estimates uncertainty for the realized date-return series $\{R_t\}$. In the empirical implementation, weights are normalized to the declared gross-exposure convention before returns are computed.

In the public equal-weight threshold implementation, optional non-negative weights $a_{t,i}$ are normalized within the selected set S_t , where S_t is produced by the declared signal and threshold rule, giving

$$\bar{r}_{t+1}(L, q, \tau) = \frac{\sum_{i \in S_t} a_{t,i} y_{t+1,i}^{(L,q)}}{\sum_{i \in S_t} a_{t,i}}. \quad (2)$$

The denominator is the total active economic weight on date t , not a generic row count. Under the primary equal-weighted evaluation $a_{t,i} = 1$, it reduces to the number of selected rows. Confidence-weighted and signal-magnitude-weighted returns are sensitivity checks.

3.2 Sharpe inference

The inference design separates estimation from testing. It reports two-sided 95 percent intervals for $\widehat{\text{SR}}$ and one-sided positive-edge p -values for $H_0 : \text{SR} \leq 0$. For a fixed

return series with positive volatility, this is equivalent to a nonpositive mean excess return null. Bootstrap inference resamples the date-return series by IID date, moving block, or stationary bootstrap. The positive-edge bootstrap p -value uses one-draw smoothing:

$$p^+ = \frac{1 + \#\{\widehat{\text{SR}}_b^* \leq 0\}}{B + 1}. \quad (3)$$

For HAC inference, reindex the finite realized date-return observations as R_u , $u = 1, \dots, T_R$, where R_u is the strategy excess return for the decision date after subtracting the applicable risk-free rate when the source is not already a long-short excess-return series. Let A_f denote the source-specific annualization factor: $A_f = 252$ for daily sources and $A_f = 12$ for monthly sources in the public registry. Let $g_u = (R_u, R_u^2)'$, $\hat{\mu} = T_R^{-1} \sum_u R_u$, $\hat{m}_2 = T_R^{-1} \sum_u R_u^2$, and $\hat{v} = \hat{m}_2 - \hat{\mu}^2$. The denominator uses the T_R -moment variance, not the $T - 1$ bias correction, because it is the plug-in moment entering the smooth functional

$$f(\mu, m_2) = \sqrt{A_f} \frac{\mu}{(m_2 - \mu^2)^{1/2}}.$$

Following Ledoit and Wolf (2008), the Sharpe ratio is treated as a smooth functional $f(\theta)$ of the moment vector $\theta = \mathbb{E}[g_t]$. On sets where $m_2 - \mu^2$ is bounded away from zero, this map is Hadamard differentiable. The asymptotic normality of $\sqrt{T_R}(\widehat{\text{SR}} - \text{SR})$ follows from the functional delta method provided the long-run covariance matrix Ω is consistently estimated by a kernel-based HAC estimator. Evaluated at $(\hat{\mu}, \hat{m}_2)$,

$$\widehat{\text{SR}} = \sqrt{A_f} \frac{\hat{\mu}}{\sqrt{\hat{v}}}, \quad \nabla f(\hat{\mu}, \hat{m}_2) = \left(\sqrt{A_f} \frac{\hat{m}_2}{\hat{v}^{3/2}}, -\frac{1}{2} \sqrt{A_f} \frac{\hat{\mu}}{\hat{v}^{3/2}} \right)'.$$

With $\widehat{\Omega}$ the Bartlett-kernel HAC estimate of the long-run covariance of g_u , the HAC-delta variance is

$$\widehat{\text{Var}}(\widehat{\text{SR}}) = \nabla f(\hat{\mu}, \hat{m}_2)' \widehat{\Omega} \nabla f(\hat{\mu}, \hat{m}_2) / T_R. \quad (4)$$

Explicitly, for centered moments $\tilde{g}_u = g_u - \bar{g}$,

$$\widehat{\Gamma}_k = T_R^{-1} \sum_{u=k+1}^{T_R} \tilde{g}_u \tilde{g}_{u-k}', \quad \widehat{\Omega} = \widehat{\Gamma}_0 + \sum_{k=1}^{K_T} \left(1 - \frac{k}{K_T + 1} \right) (\widehat{\Gamma}_k + \widehat{\Gamma}_k').$$

This is a covariance matrix for the joint process (R_u, R_u^2) , so the off-diagonal terms carry the long-run covariance between the sample mean and second moment into the Sharpe-ratio delta method. The bandwidth K_T is a tuning choice and may depend on T . The implementation reports the selected bandwidth and uses a documented automatic Bartlett bandwidth as the default; important empirical claims should be checked against reasonable nearby bandwidths and against the block-bootstrap intervals rather than relying on a single HAC lag choice. The technical appendix reports prewhitened HAC and

fixed-fraction HAC sensitivity checks.

Because financial date returns can be heavy-tailed, HAC-delta evidence is reported alongside bootstrap p -values and intervals. When the bootstrap evidence is materially weaker than the HAC evidence, the candidate is treated as robustness-fragile rather than as a robust positive-edge finding.

4 Audit protocol and implementation

4.1 Public-panel momentum implementation

The following construction is used for the public momentum-panel implementation; the sampling-boundary diagnostic itself does not depend on this particular signal. For look-back L and skip q , the skipped compounded signal is

$$s_{t,i}^{(L,q)} = \prod_{j=t-L-q+1}^{t-q} (1 + r_{j,i}) - 1. \quad (5)$$

With $q \geq 1$, the signal excludes the outcome return and is executable at date t when all signals in the available universe A_t are known before the trading decision. The signed one-period row return is

$$y_{t+1,i}^{(L,q)} = \text{sign}(s_{t,i}^{(L,q)}) r_{t+1,i}, \quad (6)$$

with $\text{sign}(0) = 0$; zero-signal and missing-signal rows receive zero economic weight. The confidence score $c_{t,i}^{(L,q)}$ is the within-date percentile rank of $|s_{t,i}^{(L,q)}|$, and the selected set is

$$\mathcal{S}_t(L, q, \tau) = \{i : c_{t,i}^{(L,q)} \geq \tau\}.$$

The percentile score is a baseline screening rule, not an optimal weighting theorem. Tie handling, one-name dates, and selected-count granularity are reported in the appendix and generated artifacts.

4.2 Researcher menus, placebos, and factor alpha

For a fixed menu $\mathcal{M}$ of lookbacks and thresholds, the primary joint adjustment is Romano-Wolf stepdown max- t resampling over $\{R_u(m) : m \in \mathcal{M}\}$. Holm-adjusted marginal bootstrap p -values and White’s Reality Check are secondary diagnostics. The Deflated Sharpe Ratio diagnostic of Bailey et al. (2014) is reported as an additional selection-bias check. In the public campaign, the menu is the finite registry-defined set

$$\mathcal{M} = \{(c, \theta_c, \tau) : c \in \mathcal{C}, \tau \in \mathcal{T}_c\},$$

where θ_c denotes the candidate’s frozen signal definition. For panel candidates, $\mathcal{T}_c = \{0.0, 0.3, 0.5, 0.7, 0.9\}$ and any lookback and skip values are fixed in `code/campaign_registry.json` rather than selected after seeing results. Pre-aggregated AQR factor-series candidates use the singleton threshold menu $\mathcal{T}_c = \{0.0\}$; because they do not contain row-level constituent signals, same-date panel permutation is not applicable to them. The reason is mechanical. Heuristically, under a Gaussian independent-strategy approximation, if N independent strategies with zero true Sharpe are searched over a sample of length T , the maximum observed daily Sharpe is on the order of $\sqrt{2\log(N)/T}$ before any economic signal exists. Real researcher menus are usually correlated rather than independent, so the procedure uses Romano-Wolf stepdown to control familywise error while preserving the joint dependence among menu elements.

The fixed-menu condition is a confirmatory convention, not a claim that research is never iterative. In practice, exploratory work should be separated from the final evaluation by freezing the signal definitions, lookbacks, thresholds, filters, cost assumptions, and benchmark model before the confirmatory run. If discarded variants influenced the final choice, they belong in the researcher menu or the exercise should be reported as exploratory. A stronger design uses a locked holdout period or an independently refreshed sample after the menu is frozen.

The public placebo is a same-date permutation design. For each date, the vector of signals is randomly permuted across portfolios, preserving the date’s cross-sectional signal distribution and missingness while breaking the signal-return pairing. The signed next-day returns, confidence ranks, selected counts, turnover, and date aggregation are then recomputed. Repeating this procedure gives a permutation null for the observed public momentum panel.

This placebo is a test under a within-date exchangeability hypothesis: under the null, the assignment of signals $s_{t,i}$ to next-period returns $r_{t+1,i}$ is exchangeable within each date t . If signals are structurally tied to size, sector, liquidity, country, exchange, beta, or other common-factor state variables, an unrestricted same-date permutation can break the wrong null. In those cases the analysis should permute within registry-defined blocks, residualize returns against the structural variables before permutation, or mark the placebo as not applicable. The permutation gate is therefore a design-specific diagnostic rather than a universal validity guarantee. If the permutation null has near-zero dispersion, the placebo is treated as uninformative for that threshold rather than as calibrated evidence for or against predictability.

The factor-alpha test estimates

$$R_t = \alpha + \beta_M(MKT_t - RF_t) + \beta_S SMB_t + \beta_H HML_t + \beta_U MOM_t + \varepsilon_t, \quad (7)$$

using HAC standard errors. The null is $H_0 : \alpha \leq 0$. Including momentum is essential

because the public signal is itself momentum-like. The alpha table is a diagnostic rather than the main gate for canonical benchmark validation.

4.3 Turnover-scaled costs

For the equal-weighted threshold portfolio, define target weights

$$w_{t,i} = \frac{\text{sign}(s_{t,i})\mathbf{1}\{i \in \mathcal{S}_t\}}{|\mathcal{S}_t|}, \quad w_{t,i} = 0 \text{ if } |\mathcal{S}_t| = 0. \quad (8)$$

Daily turnover is the half- L^1 distance between adjacent target weight vectors,

$$TO_t = \frac{1}{2} \sum_i |w_{t,i} - w_{t-1,i}|, \quad \widehat{TO} = T_{TO}^{-1} \sum_t TO_t, \quad (9)$$

where the average is over adjacent dates with at least one active side. For cost c per full rebalance in decimal units, the cost-adjusted Sharpe ratio is

$$\widehat{\text{SR}}_{net}(c) = \frac{\hat{\mu} - \widehat{TO}c}{\hat{\sigma}} \sqrt{A_f}. \quad (10)$$

This is a first-order cost stress. It assumes linear cost per unit of turnover and holds the return volatility fixed after subtracting costs. These assumptions are deliberately simple and are used only as a screening stress. The resulting cost drag should be read as a lower bound when liquidity demand is material. They are not a full execution model for high-frequency, capacity-constrained, or illiquid strategies. Such applications require spread, impact, latency, borrow, and capacity models before any tradability claim. This is a conservative cost-stress screen, not a tradability certificate. The empirical artifacts also report the break-even cost in basis points, defined by $\hat{\mu} - \widehat{TO}c = 0$. An optional square-root impact sensitivity extension in the spirit of Almgren and Chriss (2001) and the broader market-impact literature (Bouchaud et al., 2008) is provided in the technical appendix. The public sorted-portfolio examples do not contain average daily volume or order-size information, so they do not support a capacity claim.

4.4 Composite audit-gate decision

The protocol reports component diagnostics, but the final claim boundary is deliberately stricter than any single statistic. The gates are a lexicographic filter: a candidate must first clear the date-level statistical boundary, then the dependent-bootstrap robustness boundary, then the researcher-menu and placebo integrity boundaries, and finally the turnover-cost sensitivity boundary. A candidate is reported as passing the audit only if

273 it clears a registry-defined composite gate:

$$p_{\text{boot}} \leq \alpha, \quad p_{\text{perm}} \leq \alpha, \quad p_{\text{HAC}} \leq \alpha, \quad p_{\text{RW}} \leq \alpha, \quad \widehat{\text{SR}}_{\text{net}}(c) > 0.$$

274 This is a conservative pre-specified reporting policy, not a uniformly most powerful test,
 275 an optimal weighted test, or a claim that the gates exhaust all possible failure modes.
 276 The gates are conjunctive because each one addresses a different reason a performance
 277 claim can fail: bootstrap-fragile Sharpe evidence, a placebo-consistent signal, dependence-
 278 adjusted insignificance, researcher-menu selection, or nonpositive net economics. Appli-
 279 cations with different decision costs may pre-specify different gates, but they should not
 280 choose the gate sequence after seeing the results. The reported artifacts make the gate
 281 threshold reproducible: in addition to the conventional $\alpha = 0.05$, a reader can recompute
 282 pass counts at stricter levels such as $\alpha = 0.01$ without changing the statistical objects
 283 being tested. The same-date permutation gate is a panel placebo: it is applicable only
 284 when row-level signals can be permuted within date. Pre-aggregated factor series can
 285 still be economically interesting and can pass HAC-delta or Romano-Wolf tests, but they
 286 do not contain the row-level signal panel needed for this placebo. Such cases are reported
 287 as not applicable for the permutation gate rather than as passed panel evaluations.

288 The stages have different interpretations. The permutation gate asks whether the
 289 signal-return pairing beats a same-date placebo. The HAC gate asks whether the re-
 290 sulting date-return Sharpe is positive after serial dependence and heteroskedasticity are
 291 accounted for. The bootstrap gate asks whether that conclusion survives resampling of
 292 the date-return series. The Romano-Wolf gate asks whether the claim survives the fixed
 293 researcher menu. The cost gate asks whether the edge remains economically usable after
 294 turnover-scaled frictions. Factor-alpha, holdout, subperiod, and stationarity diagnostics
 295 are reported alongside these gates to prevent a single favorable statistic from becoming
 296 the claim.

297 5 Sharpe UVIF theory

298 5.1 Mean-return special case

299 **Assumption 1** (Equicorrelated row model). *For each date t , selected row returns satisfy*

$$y_{t,i} = \mu + \lambda_t + \epsilon_{t,i},$$

300 *where λ_t is a common date shock, $\epsilon_{t,i}$ is idiosyncratic, λ_t is uncorrelated with $\epsilon_{t,i}$,*
 301 *$\text{Var}(\lambda_t) = \sigma_\lambda^2$, and $\text{Var}(\epsilon_{t,i}) = \sigma_\epsilon^2$. The selected count is $n_t = \bar{n}$ for the displayed*
 302 *derivation, and dates are independent.*

Under this model,

$$\sigma^2 = \text{Var}(y_{t,i}) = \sigma_\lambda^2 + \sigma_\epsilon^2, \quad \rho = \text{Corr}(y_{t,i}, y_{t,j}) = \frac{\sigma_\lambda^2}{\sigma_\lambda^2 + \sigma_\epsilon^2} \quad (i \neq j).$$

Thus ρ is the intraclass correlation induced by the common date shock; economically, it is the fraction of row-return variance explained by shocks shared by same-date selections. In empirical diagnostics, ρ is estimated after the candidate threshold is fixed. For each panel candidate, the selected signed row returns are aligned by date and asset or portfolio, and $\hat{\rho}$ is the average off-diagonal sample correlation among active row-return series with overlapping dates; $\bar{n}$ is the average selected count across active dates. When the selected set varies over time, this equicorrelation estimate is a descriptive compression of the same-date covariance structure, not an identifying assumption for the confirmatory HAC or bootstrap tests.

Proposition 1 (Mean-return design effect). *Under the equicorrelated row model, the ratio of the naive row-pooled reported standard error of the mean to the date-level standard error is*

$$\frac{se_{row,naive}}{se_{date,true}} = \{1 + (\bar{n} - 1)\rho\}^{-1/2}.$$

For $\rho > 0$ and $\bar{n} > 1$, the naive row-level standard-error report is too small.

This calculation is not new as a sampling result; it is the Moulton-like special case of the paper’s UVIF logic, written for signed trading-panel rows and date-level portfolio returns. To avoid conflict with the researcher menu notation $\mathcal{M}$, write the cross-sectional variance design effect as

$$\text{UVIF}_{\mu, xs}(\bar{n}, \rho) = 1 + (\bar{n} - 1)\rho.$$

The square root is reported when the object is a standard-error or t -statistic adjustment:

$$\text{UVIF}_{\mu, xs}^{1/2}(\bar{n}, \rho) = \sqrt{1 + (\bar{n} - 1)\rho}.$$

Thus UVIF denotes a variance ratio, while $\text{UVIF}^{1/2}$ denotes the associated standard-error multiplier. This mean-return calculation is expository: it makes the scale of understatement visible before the paper turns to HAC and bootstrap decisions on the realised date-return series. The magnitude can be economically large. The relevant inflation factor for the true standard error relative to the naive row report is $\text{UVIF}_{\mu, xs}^{1/2}(\bar{n}, \rho)$. With $\bar{n} = 50$, increasing same-date correlation from $\rho = 0.10$ to $\rho = 0.60$ raises this factor from 2.43 to 5.51. Thus the largest damage occurs precisely in regimes where many names are selected and common shocks dominate idiosyncratic noise, which is the pattern one expects during market stress.

The proof is given in the technical appendix.

Remark 1 (Heterogeneous factor covariance). *The equicorrelated model is a closed-form illustration, not a restriction imposed by the protocol. Let a same-date signed-return vector satisfy*

$$y_t = \mu \mathbf{1} + B f_t + u_t,$$

where B contains asset-specific loadings, $\text{Var}(f_t) = \Sigma_f$, and $\text{Var}(u_t) = D_t$. Then the same-date covariance matrix is

$$\Sigma_t = B \Sigma_f B' + D_t,$$

with heterogeneous pairwise correlations whenever loadings or idiosyncratic variances differ. For any displayed portfolio weights w_t , the relevant one-date variance is $w_t' \Sigma_t w_t$, not the row-independent variance. The protocol avoids estimating the full Σ_t as a prerequisite for inference by moving the evidence to the realised date-return series and then using HAC or dependent bootstrap inference on that series. The design-effect calculation is therefore best read as the transparent special case that explains why row pooling fails.

5.2 Unified variance inflation for the Sharpe ratio

The mean-return design effect is useful but incomplete because the reported performance metric is usually the Sharpe ratio. The Sharpe ratio is a nonlinear functional of the first two moments, so its variance inflation depends on the covariance of R_t and R_t^2 , not only on the variance of R_t . This motivates the Unified Variance Inflation Factor:

$$\text{UVIF}_{\text{SR}} = \frac{\text{Var}_{LR}(\widehat{\text{SR}}_{\text{portfolio}})}{\text{Var}_{\text{naive}}(\widehat{\text{SR}}_{\text{row}})}. \quad (11)$$

The numerator is the long-run delta-method variance at the date-return boundary. The denominator is the row-pooled delta-method variance that would be reported if the selected rows were treated as independent observations. Equation (11) is therefore a reported-uncertainty distortion ratio. It compares the uncertainty of the economically relevant date-level Sharpe claim with the uncertainty that would be reported under row-pooled IID analysis. Under exchangeability of the linearized Sharpe influence function, and when the row-pooled and date-level estimators target the same equal-weight Sharpe functional, the cross-sectional part of this variance ratio reduces to the same $1 + (\bar{n} - 1)\rho$ design effect shown above, with ρ applied to the linearized Sharpe influence rather than to raw returns. Outside that special case, Sharpe UVIF is target-aware: it measures the gap between the row-level report and the date-level claim that can be traded, allocated to, or interpreted as portfolio evidence.

Assumption 2 (Sharpe UVIF regularity). *The date-level process $G_t = (R_t, R_t^2)'$ is stationary with finite $4 + \delta$ moments, $v_R = m_{2,R} - \mu_R^2 \geq \eta > 0$, and finite long-run covariance*

Ω_R . The aggregation rule is fixed before evaluation. The displayed decomposition uses a constant selected-count approximation $\bar{n}$, and the row-pooled IID denominator is a counterfactual reporting benchmark rather than the maintained sampling law.

Proposition 2 (Sharpe UVIF decomposition). Let R_t be the date-level portfolio excess return and $G_t = (R_t, R_t^2)'$. Define $\theta_R = (\mu_R, m_{2,R})' = \mathbb{E}[G_t]$, $v_R = m_{2,R} - \mu_R^2 > 0$. For source-specific annualization factor A_f , define

$$d_R = \nabla f(\theta_R) = \left(\sqrt{A_f} \frac{m_{2,R}}{v_R^{3/2}}, -\frac{1}{2} \sqrt{A_f} \frac{\mu_R}{v_R^{3/2}} \right)'.$$

Let $\Gamma_k^R = \text{Cov}(G_t, G_{t-k})$ and $\Omega_R = \Gamma_0^R + \sum_{k \geq 1} (\Gamma_k^R + \Gamma_k^{R'})$. For the selected row return $Y_{t,i}$, let $H_{t,i} = (Y_{t,i}, Y_{t,i}^2)'$, $\theta_Y = \mathbb{E}[H_{t,i}]$, $d_Y = \nabla f(\theta_Y)$, and $\Psi_Y = \mathbb{V}\text{ar}(H_{t,i})$. With a constant selected count $\bar{n}$, the first-order Sharpe variance inflation relative to a row-pooled IID report is

$$\text{UVIF}_{\text{SR}} = \underbrace{\frac{\bar{n} d_R' \Gamma_0^R d_R}{d_Y' \Psi_Y d_Y}}_{\text{cross-sectional UVIF component}} \times \underbrace{\frac{d_R' \Omega_R d_R}{d_R' \Gamma_0^R d_R}}_{\text{Sharpe long-run serial factor}}. \quad (12)$$

Moreover, the serial factor can be written as

$$1 + 2 \sum_{k=1}^{\infty} \psi_k^{\text{SR}}, \quad \psi_k^{\text{SR}} = \frac{d_R' \Gamma_k^R d_R}{d_R' \Gamma_0^R d_R},$$

and the scalar long-run variance entering the numerator is

$$d_R' \Omega_R d_R = \frac{A_f}{v_R^3} \left[m_{2,R}^2 \Omega_{11} - \frac{\mu_R m_{2,R}}{2} (\Omega_{12} + \Omega_{21}) + \frac{\mu_R^2}{4} \Omega_{22} \right], \quad (13)$$

where Ω_{ab} denotes the (a, b) element of Ω_R .

The proof is given in the technical appendix.

5.3 What Proposition 2 does and does not identify

Proposition 2 is a first-order variance decomposition for a fixed aggregation rule and a constant selected-count approximation. It identifies how a row-pooled IID Sharpe uncertainty report changes when the claim is moved to the date-return boundary and when the date series has long-run serial covariance in (R_t, R_t^2) . It does not identify a causal trading edge, does not prove that row and date Sharpe functionals are equal outside the stated exchangeability conditions, and does not replace the confirmatory date-level HAC, bootstrap, researcher-menu, placebo, factor-alpha, and cost diagnostics. The empirical role of Sharpe UVIF is therefore diagnostic: it explains the sampling-boundary distortion that the claim-boundary diagnostics then evaluate with date-level

inferential procedures. The sample UVIF reported in the empirical section is a plug-in diagnostic for this population ratio; confirmatory rejection decisions do not depend on estimating UVIF consistently.

Remark 2 (Connection to Moulton-like design effects). *If the linearized row Sharpe influence $\zeta_{t,i}^Y = d_Y'(H_{t,i} - \theta_Y)$ is exchangeable within date with intraclass correlation ρ_ζ , and the date-level linearization is the equal-weighted average of those row influences, the cross-sectional factor in (12) reduces to $1 + (\bar{n} - 1)\rho_\zeta$. The mean-return design effect is the special case in which the influence variable is $Y_{t,i} - \mu$. The Sharpe case replaces raw return correlation by correlation of the linearized nonlinear-performance influence, which is why skewness, kurtosis, and volatility clustering enter the variance inflation.*

5.4 Higher-moment UVIF and non-normality

The two-moment Sharpe UVIF already depends on fourth moments through the variance of R_t^2 . If a venue or application uses an adjusted Sharpe ratio that explicitly corrects for skewness and kurtosis, the same delta-method construction extends the moment vector from $(R_t, R_t^2)'$ to $(R_t, R_t^2, R_t^3, R_t^4)'$. The resulting higher-moment UVIF is the ratio of the adjusted-Sharpe long-run variance at the date-return boundary to the corresponding row-pooled IID variance. It is reported as a tail-risk diagnostic, not as an additional rejection gate. The full expression is in the technical appendix.

5.5 Effective row information retention

For interpretability, the paper reports Effective Row Information Retention (ERIR) as a design-effect diagnostic rather than as an additional hypothesis test or formal information claim. Under the equicorrelated row model, the cross-sectional mean-return design effect is

$$DE_{\mu, xs} = 1 + (\bar{n} - 1)\rho. \quad (14)$$

The corresponding row-retention diagnostic is therefore

$$\text{ERIR} = \{1 + (\bar{n} - 1)\rho\}^{-1}. \quad (15)$$

ERIR is interpreted as the fraction of nominal row-level information that remains nonredundant under the equicorrelation approximation. For example, $\text{ERIR} = 0.111$ means that approximately $1 - 0.111 = 0.889$, or 89 percent, of the nominal row-level information implied by naive row pooling is redundant under this diagnostic approximation. The statistic is descriptive only. Confirmatory inference is conducted using date-level HAC-delta and dependent-bootstrap procedures.

416 **Assumption 3** (Fixed-menu bootstrap validity). *For each fixed menu element m , the*
417 *date-return series $\{\bar{r}_t(m)\}$ is strictly stationary and alpha-mixing with moments of order*
418 *$4 + \delta$. The long-run covariance of $(\bar{r}_t, \bar{r}_t^2)$ is finite and non-degenerate, and $\mathbb{V}\text{ar}(\bar{r}_t) \geq \eta >$*
419 *0. The block length satisfies $\ell \rightarrow \infty$ and $\ell/T \rightarrow 0$. The researcher menu $\mathcal{M}$ is fixed and*
420 *finite.*

421 **Corollary 1** (Fixed-menu validity statement). *Under the fixed-menu bootstrap-validity*
422 *assumption, the HAC-delta and block-bootstrap procedures consistently estimate the*
423 *marginal sampling law of $\widehat{\text{SR}}(m)$ for each fixed m . If marginal positive-edge p -values are*
424 *valid and the stepdown rejection regions are monotone, Holm and Romano-Wolf stepdown*
425 *control familywise error asymptotically for the fixed menu.*

426 **Justification.** This is an assembly of standard results rather than a new theorem. The
427 Sharpe functional is Hadamard differentiable on the set $m_2 - \mu^2 \geq \eta$. A central limit
428 theorem for the sample mean of $(\bar{r}_t, \bar{r}_t^2)$, the delta method, and standard block-bootstrap
429 consistency for mixing sequences imply marginal Sharpe validity (Kunsch, 1989; Politis
430 and Romano, 1994; Lahiri, 2003). For a fixed finite menu, Holm controls FWER for valid
431 marginal p -values, and Romano-Wolf stepdown uses joint max-statistic resampling under
432 the standard monotonicity requirement (Holm, 1979; Romano and Wolf, 2005).

433 **Remark 3** (Scope of the validity statement). *The validity statement does not cover long*
434 *memory, infinite-variance heavy tails, growing researcher menus, structural breaks, or*
435 *nonstationary date returns. The empirical evaluation reports stationarity diagnostics and*
436 *stress simulations, but those outputs are caveats and diagnostics, not proof of asymptotic*
437 *validity in those settings.*

438 6 Synthetic positive-control validation

439 The claim boundary is not mechanically anti-discovery. To check that the composite
440 gates can pass a true signal, a synthetic panel is generated with a persistent predictive
441 signal, moderate same-date dependence, and serial dependence in the date-level returns.
442 The signal is constructed before inference is run, the same date-aggregation and audit
443 gates are applied, and the resulting positive-control candidate passes HAC, Romano-Wolf,
444 and same-date permutation checks. This validation is not used as evidence for a public
445 trading strategy; it shows that the procedure can accept known positive evidence when
446 the data-generating process contains a genuine date-level edge.

447 The positive-control DGP uses $T = 5000$ dates and $N = 100$ entities, annual alpha
448 0.12 (daily alpha 0.12/252), idiosyncratic volatility 0.01, common-factor loading 0.50,
449 AR(1) common-factor coefficient 0.10, signal-stay probability 0.98, and seed 777. The
450 generated artifact records the bootstrap and permutation counts used for the run. The

design is intentionally simple: it is a known-edge calibration check for the audit gate, not a model of any public anomaly.

Table 1: Synthetic positive-control evaluation.

Panel	$\bar{\rho}$	Adj.	SR	HAC p^+	RW p	Perm. p^+
Synthetic control	0.202	4.58	1.744	7.67e-13	2.00e-04	1.00e-04

Appendix material. The worked numerical example, full proof details, positive-control data generation notes, and secondary robustness artifacts are reported in the technical appendix. The main manuscript keeps only the claim boundary and the evidence needed to evaluate it.

7 Public benchmark and stress-test evidence

The public experiment is a registry-defined candidate campaign rather than a single example. It evaluates the canonical Kenneth French winner-minus-loser momentum decile benchmark, a broader Kenneth French decile-rank threshold profile, Kenneth French 25 Size/BM portfolios with a skipped-momentum signal, and public AQR value, momentum, quality, beta, and HML-style factor candidates. The AQR series are economically meaningful factor-series benchmarks, but they are pre-aggregated series rather than row-level constituent panels; when the same-date permutation gate cannot be applied, the benchmark is outside the composite panel-audit gate. The main tables therefore separate panel candidates from single-series factor benchmarks. Failed public-data access attempts are kept in the provenance artifacts rather than treated as main empirical results.

The generated empirical tables in this section are produced from a full-campaign root by `code/render_manuscript_artifacts.py`. Table 2 reports panel candidates subject to the composite audit gate. Table 3 reports pre-aggregated factor benchmarks subject to time-series inference only. Table 4 verifies that the canonical WML candidate matches the direct French winner-minus-loser decile benchmark. Table 5 gives the dependence audit summary using row p -values, UVIF, ERIR, audit p -values, and net Sharpe status for panel candidates. Table 6 reports the targeted $L = 1, 5, 21$ horizon-effect audit for the dynamic Size/BM momentum panel. The technical appendix reports the full secondary table set: sampling-boundary diagnostics, same-date permutation details, HAC bandwidth/prewhitening/fixed- b robustness, gate sensitivity, hold-out splits, and cost sweeps. The production metadata include the SHA256 hash of `code/campaign_registry.json`; this hash is the reproducibility record for the frozen public sources, candidate types, source frequencies, annualization factors, thresholds, lookbacks, skips, and eligibility rules used in the rendered campaign.

The factor-alpha, data-snooping, stationarity, holdout, and cost tables are secondary diagnostics, not replacements for the primary date-return decision. A raw positive Sharpe that does not survive FF3 plus momentum alpha, bootstrap robustness, Romano-Wolf/White checks, same-date placebos, or plausible costs should not be interpreted as reliable portfolio-performance evidence. This is the empirical implication of the claim-boundary design: familiar public candidates can look attractive at one boundary and fail at the composite decision boundary.

The panel-candidate table reports only sources with row-level signal-return structure and therefore eligibility for the composite panel audit gate. The single-series benchmark table reports pre-aggregated factor returns separately; those rows are time-series benchmarks, not failed panel-audit candidates.

Table 2: Panel candidates subject to the composite audit gate.

Panel candidate	SR	HAC p+	Boot p+	RW p	Perm p	Net SR	Audit gate 5%
French momentum deciles, rank-threshold panel	0.447	7.54e-05	2.00e-04	2.00e-04	1.000	0.447	no
French WML decile benchmark	0.497	5.53e-06	2.00e-04	2.00e-04	1.00e-04	0.497	yes
French 25 Size/BM, skipped-momentum panel	0.058	0.412	0.422	0.480	0.943	-0.050	no

Table 3: Single-series factor benchmarks subject to time-series inference only.

Single-series benchmark	SR	HAC p+	Boot p+	RW p	Scope
AQR Betting Against Beta, equity factors	0.741	3.08e-13	2.00e-04	2.00e-04	time-series only
AQR Quality Minus Junk	0.536	2.26e-05	2.00e-04	2.00e-04	time-series only
AQR HML Devil	0.336	0.004	0.002	0.002	time-series only
AQR Value and Momentum Everywhere	0.572	1.62e-04	2.00e-04	2.00e-04	time-series only

Note: Time-series only means not eligible for same-date permutation, ERIR, or the composite panel audit gate.

The benchmark validation table shows that the panel implementation matches the direct French winner-minus-loser construction.

Table 4: Canonical French momentum benchmark validation.

Candidate	thr	n	start	end	panel SR	direct WML SR	scale
French WML decile benchmark	0.900	26131	1926-11-04	2026-04-30	0.497	0.497	0.500

The dependence audit table summarizes the boundary directly: row-level evidence can disappear once same-date redundancy, ERIR, placebo evidence, and turnover-scaled net Sharpe are evaluated together.

The horizon table shows that changing the lookback reduces turnover in the Size/BM stress panel but does not convert it into a robust net-Sharpe claim.

The campaign separates benchmark validation from broader search claims. The canonical French WML candidate is expected to match the direct Kenneth French winner-minus-loser decile construction and is reported as a public benchmark check. Its audit-gate pass does not undermine the sampling-boundary argument; it shows that the claim

Table 5: Dependence audit summary for panel candidates.

Candidate	Row p+	UVIF	ERIR	Audit p+	Net SR	Status
French momentum deciles, rank-threshold panel	1.70e-04	1.000	1.000	1.000	0.447	Fails permutation gate
French WML decile benchmark	4.85e-04	1.000	1.000	2.00e-04	0.497	Passes composite gate
French 25 Size/BM, skipped-momentum panel	0.253	9.002	0.111	0.943	-0.050	Fails statistical and cost gates

Note: ERIR is a design-effect retention diagnostic and is not used as a rejection rule.

Table 6: Horizon-effect audit for the dynamic Size/BM momentum panel.

L	thr	dates	avg names	gross SR	net SR 5bps	turnover	UVIF SR	serial UVIF	HAC p+
1	0.500	3497	12.913	0.073	-0.394	0.733	11.498	0.891	0.386
5	0.500	3493	13.000	-0.312	-0.528	0.339	12.880	0.990	0.878
21	0.500	3477	13.000	0.058	-0.050	0.169	12.121	0.933	0.412

boundary can preserve canonical date-level evidence when the signal is strong enough and the placebo, menu, dependence, and cost diagnostics do not overturn it. This resolves the apparent $\widehat{SR} = 0.082$ benchmark problem in earlier artifacts: that number came from an earlier rank-threshold stress candidate, not from the canonical WML construction. In the rendered campaign, the WML panel Sharpe matches the direct WML Sharpe at approximately 0.50, and the scale difference is exactly one-half because the panel portfolio holds equal long and short legs while the direct WML series is winner minus loser. The broader rank-threshold profile and the French Size/BM dynamic-momentum panel are evaluated as stress cases rather than as canonical anomaly replications. The AQR factor-series candidates have positive time-series evidence in the reported HAC and bootstrap diagnostics, but because they are pre-aggregated factor returns rather than row-level signal panels, they cannot be promoted to passed panel-evaluation candidates.

The horizon-effect audit shows that the dynamic Size/BM momentum stress case fails because of weak signal and execution drag, not because the canonical momentum benchmark is missing. At threshold 0.5, $L = 1$ has very high turnover and a negative 5 bps net Sharpe; $L = 21$ lowers turnover substantially but still leaves net Sharpe slightly negative. The empirical Sharpe UVIF relative to row-naïve inference is about 11–13 across the three horizons, while the serial-only UVIF is close to one in this sample. Thus the dominant inflation channel in this public stress panel is cross-sectional row multiplication; a longer lookback does not mechanically make serial dependence the dominant force. For the dynamic Size/BM panel in Table 5, the reported ERIR is 0.111. Under the equicorrelation diagnostic, this means $1 - 0.111 = 0.889$, so approximately 89 percent of the row-level information implied by naïve row pooling is redundant. The dependence audit summary is intentionally artifact-backed rather than illustrative. It does not assert that every named factor is a panel candidate. Pre-aggregated AQR factor series are excluded from that table because they do not provide row-level signal-return pairs from which row p -values, same-date permutation tests, or panel ERIR can be computed.

8 Monte Carlo evidence

The Monte Carlo study is designed as a size-and-power check of the sampling boundary, not as a search for a favorable strategy. Each replication generates a symbol-date return panel, applies the same date-aggregation rule used in the empirical analysis, and then asks whether each inference method rejects a positive-edge null at the nominal 5 percent level. The base panel is

$$r_{t,i} = \mu_t + \beta_i' f_t + \epsilon_{t,i}, \quad f_t = \Phi f_{t-1} + u_t.$$

The grid varies serial dependence, cross-sectional dependence, GARCH volatility, multifactor exposure, and regime switching. The high-persistence AR(1) case is reported as such; it is not labeled as a long-memory design. The high-volatility GARCH case is not described as an infinite-variance heavy-tail design. The production campaign uses 1000 Monte Carlo replications for the displayed size and rejection-rate tables, $T = 1000$ dates and $N = 50$ entities in the main null designs, 5000 bootstrap draws per simulation, and seed-indexed replications generated from the simulation index. Moving-block and stationary bootstrap procedures use the same block-length rule as the empirical audit code, and the full DGP grid, seed policy, registry hash, and run metadata are included in the reproducibility package. The null designs set the true edge to zero while varying cross-sectional correlation and serial memory; the power designs then increase the true annualized Sharpe on the same dependence-aware decision boundary. The important object is the gap between the row-naive rejection rate and the rejection rates of methods that operate on date-level returns. In the displayed null grid, the row-naive test overrejects around 35–37 percent under dependent nulls, while HAC-delta, moving-block bootstrap, stationary bootstrap, and Romano-Wolf remain much closer to nominal size. In the power grid, row-naive rejection reaches about 50 percent at a true annualized Sharpe near 0.19, before the dependence-aware methods show comparable power; that gap is false precision, not usable evidence.

The size and power experiment compares row-level naive testing, date-IID bootstrap, HAC-delta inference, moving-block bootstrap, stationary bootstrap, and Romano-Wolf joint resampling. The coverage experiment focuses on the date-level Sharpe target, so row-naive intervals are interpreted as coverage of the economically relevant date-level claim rather than coverage of a separate row-distribution Sharpe. The purpose is not to rank methods by power alone. A method that appears powerful because it overrejects under the null is not acceptable for confirmatory evaluation.

Table 7 reports the compact simulation and resampling settings used for the rendered campaign. Table 8 gives the main size evidence. The row-naive column is the familiar symbol-date analysis; the HAC-delta, moving-block, stationary, and Romano-Wolf

columns make decisions from the date-return series. When row-naive rejection is high in null or near-null designs but the dependence-aware methods remain near nominal size, the apparent extra power is a false precision gain from treating same-date rows as independent evidence. Table 9 reports row-naive interval behavior against the date-level Sharpe target, making the target mismatch explicit. Table 10 then sweeps same-date correlation and selected-count pressure under a zero-edge null. These sweeps are diagnostic stress tests for the sampling boundary; they do not introduce a new empirical claim. The technical appendix reports the full DGP grid, coverage table, power table, and power-curve figure.

The null-size figure and table are the core simulation results: under dependent nulls, row-naive testing rejects about one third of the time at a nominal 5 percent level, while date-level HAC, block-bootstrap, stationary, and Romano-Wolf procedures stay much closer to size.

Table 7: Monte Carlo and resampling settings.

Setting	Value
Monte Carlo replications	1000
Bootstrap draws	5000
Permutation draws	10000
Main null panel sizes	T=1000, N=50
Cross-sectional correlation grid	0.000, 0.200, 0.250, 0.300, 0.350, 0.400, 0.500
AR(1) grid	0.000, 0.150, 0.200, 0.300, 0.500, 0.700
GARCH parameter pairs	(0.050, 0.900); (0.100, 0.850); (0.150, 0.800)
HAC bandwidth rule	automatic Bartlett, with fixed-lag sensitivity checks
Moving-block length	ceil(T to the 1/5 times max(D,1)), capped at floor(T/3)
Stationary mean block	same length as moving-block rule
Dependence factor D	(2 rho1/(1-rho1)) to the 2/3 for positive AR(1) rho1; otherwise 1
Seed policy	seed-indexed replications from campaign runner

Table 8: Null rejection rates at nominal 5 percent size.

dgp	date_iid	hac_delta	moving_block	romano_wolf	row_naive	stationary
null_dep	0.094	0.066	0.070	0.068	0.352	0.067
null_garch	0.098	0.063	0.070	0.067	0.351	0.068
null_iid	0.060	0.060	0.061	0.060	0.062	0.062

9 Discussion and limitations

The results should be read as a financial-econometrics testing exercise for entity-time panel evidence. The framework addresses two common sources of false precision: same-date cross-sectional row multiplication and serial dependence in the date-level return

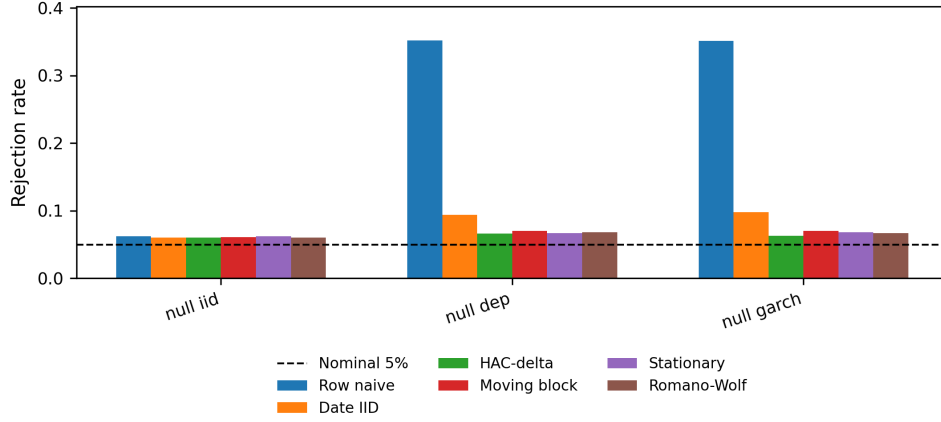


Figure 1: Null rejection rates under dependent and IID designs. The dashed line is the nominal 5 percent rejection rate.

Table 9: Row-naive interval behavior against the date-level Sharpe target.

DGP	Name	row coverage	bias vs date SR	CI width	mean SE	n
01_iid	IID baseline	0.000	-2.567	0.278	0.071	1000
02_ser_mild	Serial mild	0.000	-2.567	0.278	0.071	1000
03_ser_strong	Serial strong	0.000	-2.568	0.278	0.071	1000
04_xs_mild	Cross mild	0.073	-0.474	0.278	0.071	1000
05_xs_strong	Cross strong	0.262	-0.153	0.278	0.071	1000
06_both_mod	Both moderate	0.088	-0.503	0.278	0.071	1000
07_both_strong	Both strong	0.178	-0.145	0.278	0.071	1000
08_realistic	Realistic	0.107	-0.277	0.124	0.032	1000
09_garch_mild	GARCH mild	0.110	-0.473	0.278	0.071	1000
10_garch_strong	GARCH strong	0.199	-0.268	0.278	0.071	1000
11_multifactor	Multi-factor K=3	0.225	-0.327	0.278	0.071	1000
12_regime	Regime-switching	0.306	-0.092	0.197	0.050	1000
13_high_ar1	High-persistence AR(1)	0.137	-0.338	0.278	0.071	1000
14_garch_highvol	High-volatility GARCH	0.193	-0.325	0.278	0.071	1000

Table 10: Target-boundary design sweeps over same-date correlation and selected-count pressure.

design	method	rej.	SE	bias	mean interval SE	N	rho	n
rho=0.3, N=50	row_naive	0.354	0.015	3.82e-04	0.071	50	0.300	1000
rho=0.3, N=50	hac_delta	0.044	0.006	9.09e-05	0.599	50	0.300	1000
rho=0.1, N=50	row_naive	0.310	0.015	0.011	0.071	50	0.100	1000
rho=0.1, N=50	hac_delta	0.064	0.008	0.031	0.598	50	0.100	1000
rho=0.5, N=50	row_naive	0.409	0.016	0.004	0.071	50	0.500	1000
rho=0.5, N=50	hac_delta	0.047	0.007	0.005	0.597	50	0.500	1000
rho=0.0, N=50	row_naive	0.097	0.009	0.002	0.071	50	0.000	1000
rho=0.0, N=50	hac_delta	0.056	0.007	0.017	0.598	50	0.000	1000
N=5, rho=0.35	row_naive	0.192	0.012	-0.013	0.225	5	0.350	1000
N=5, rho=0.35	hac_delta	0.058	0.007	-0.018	0.598	5	0.350	1000
N=25, rho=0.35	row_naive	0.308	0.015	-0.016	0.100	25	0.350	1000
N=25, rho=0.35	hac_delta	0.052	0.007	-0.026	0.600	25	0.350	1000
rho=0.7, N=50	row_naive	0.408	0.016	0.004	0.071	50	0.700	1000
rho=0.7, N=50	hac_delta	0.060	0.008	0.004	0.599	50	0.700	1000
N=50, rho=0.35	row_naive	0.368	0.015	-1.05e-04	0.071	50	0.350	1000
N=50, rho=0.35	hac_delta	0.071	0.008	-5.58e-05	0.597	50	0.350	1000
N=100, rho=0.35	row_naive	0.421	0.016	-0.002	0.050	100	0.350	1000
N=100, rho=0.35	hac_delta	0.056	0.007	-0.003	0.598	100	0.350	1000

series. It does not remove the need for economic interpretation, out-of-sample validation, liquidity analysis, or venue-specific execution modeling.

The formal validity statement assumes a fixed and finite researcher menu, stationary date-return series, adequate moments for the Sharpe delta-method expansion, and block-bootstrap regularity. The framework does not cover long-memory processes, infinite-variance returns, growing researcher menus, structural breaks, or nonstationary date returns without additional assumptions. The empirical stationarity, holdout, factor-alpha, placebo, and cost outputs are therefore diagnostics and boundary checks, not proofs that every maintained assumption holds in every market setting. A locked holdout period or independently refreshed sample after the researcher menu is frozen remains the stronger design when such data are available.

10 Data and code availability

The reproducibility package contains the LaTeX source, generated public aggregate artifacts, public-data loaders, simulation code, unit tests, release scripts, and SHA256 provenance. The public repository is: <https://github.com/DsChauhan08/dependence-aware-evaluation-of-panel-trading-strategy-evidence>

Code is released under Apache-2.0 and manuscript/public aggregate artifacts under CC-BY 4.0 unless a target venue requires different terms. Raw third-party data are not redistributed where source terms do not permit redistribution; the repository instead provides documented loaders, source links, access metadata, generated aggregate artifacts, and checksums needed to reproduce the analysis. Proprietary predictions, raw

trades, private tickers, production feature definitions, and unrelated workspace artifacts are excluded. The technical appendix contains the full simulation grid, robustness tables, positive-control artifact, registry notes, and proof details omitted from the main manuscript for length.

11 Conclusion

This paper formalizes sampling-boundary distortion in Sharpe-ratio evidence from financial panels. The resulting claim boundary moves selected symbol-date rows to date-level portfolio returns, estimates Sharpe uncertainty with HAC and dependent-bootstrap methods, corrects the fixed researcher menu, checks same-date placebo and factor-alpha boundaries, and reports turnover-scaled cost sensitivity. The Sharpe UVIF converts the familiar design-effect problem into a variance ratio for the nonlinear Sharpe functional, and ERIR reports how much nominal row-level information remains nonredundant under the equicorrelation diagnostic approximation. For researchers and allocators, the practical implication is direct: statistical significance earned by multiplying dependent rows is not by itself portfolio-performance evidence.

The public campaign shows how this claim boundary changes interpretation. The canonical French WML candidate validates the pipeline against the direct winner-minus-loser construction. By contrast, broader rank-threshold and dynamic Size/BM panel claims fail dependence-aware, placebo, researcher-menu, or cost boundaries; in the Size/BM stress panel, an ERIR of 0.111 implies that about 89 percent of nominal row-level information is redundant under the diagnostic approximation. Pre-aggregated AQR factor series remain useful time-series benchmarks, but without row-level constituent signals they cannot clear the panel permutation gate.

The resulting procedure is therefore a reproducible testing boundary rather than a strategy generator. It helps separate benchmark validation from exploratory search and makes the cost of dependence explicit before a result is promoted as performance evidence. Future work should extend the framework to nonlinear market-impact and capacity models, time-varying cross-sectional dependence, and richer blocked-permutation designs.

Author contributions

Dhananjay S. Chauhan is the sole author of this manuscript and was responsible for conceptualization, methodology, software, validation, formal analysis, data curation, writing, and revision.

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Conflict of interest

The author declares no conflict of interest.

Ethics statement

This study uses publicly available aggregate financial data and simulation data. It does not involve human participants, personal data, clinical data, animal subjects, or cell lines. Ethics approval was therefore not required.

Data availability

The data used in this study are derived from publicly available financial datasets, including Kenneth French data and AQR factor data. Cleaned public aggregate artifacts, source metadata, checksums, and reproduction instructions are available at: <https://github.com/DsChauhan08/dependence-aware-evaluation-of-panel-trading-strategy-evidence>

Raw third-party data are not redistributed where source terms do not permit redistribution.

Code availability

The code used to generate the empirical results, Monte Carlo simulations, tables, figures, and manuscript artifacts is available at: <https://github.com/DsChauhan08/dependence-aware-evaluation-of-panel-trading-strategy-evidence>

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